

Artificial Intelligence (AI2002)

Date: July 15th 2025

Course Instructor(s)

Ms. Mahzaib Younas

Sessional Exam

Total Time(Hrs): 2

Total Marks: 7

Total Questions: 80

Roll No _____ Section _____

Student Signature _____

Vetted by: _____ Signature: _____

CLO#	1	3	3	3	2	2	1
Q#	1	2	3	4	5	6	7
Total Marks	08	12	10	15	10	15	10
Obtained Marks							

Attempt all the questions.

CLO [C1]: To introduce the notion of intelligence and the so called artificiality associated with it, and how these can be modelled in computational system

Q1: Agents and Environments

[4 + 4 = 8 Marks]

Part A: Read the following Scenarios carefully and answer which agent (only 1 agent per scenario) will do the following tasks.

1. A medical diagnosis assistant uses historical patient records and symptoms to generate possible diagnoses. It keeps track of symptom progression.	Learning agent
2. An AI chatbot replies based only on the most recent user message using a fixed rule-based system, without storing any history or context.	Reflex agent
3. A drone avoids birds mid-flight using pre-programmed rules based only on immediate sensor data.	Reflex agent
4. An AI lawyer weighs multiple laws and client goals to generate the best legal strategy.	Goal based agent
5. A language app adapts to the student's strengths and weaknesses and customizes the next lessons accordingly.	Learning
6. A digital assistant picks restaurant based on cuisine, price, user reviews, and travel time.	Utility

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7. A parking assistant remembers previous failed parking spots to avoid them in future attempts.	Learning
8. An AI tutor aims to help a student reach a certain grade and selects exercises that lead to that goal.	Goal

Part B: A student is developing a robot that can roll a die. The robot grips and releases the die with an arm that can also twist, and uses two cameras that can see the entire rolling area at once for feedback. The goal is to develop methods that can roll the die in a way that increases the chance of landing on a desired side of the die. Please consider the environment that is the robot, die, and rolling area. Is this environment (justify your answers!):

[4 Marks]

Environment	Yes/No	Justification
Deterministic		
Episodic		
Static		
Discrete		

CLO [C3]: Knowledge of method of blind as well as informed search and ability to practically apply the corresponding techniques

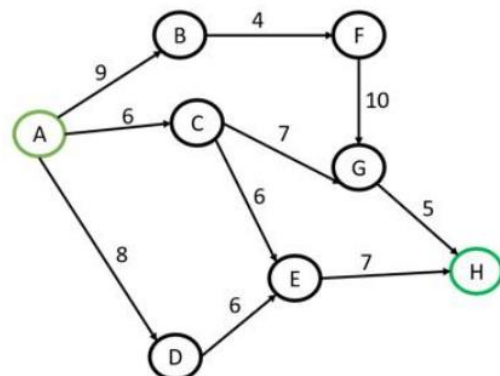
Q2: Informed Searches

[7 + 5 = 12 Marks]

Consider the graph with A as the start node and H as the goal node with heuristic values given in the table.

[marks]

A	19
B	20
C	11
D	13
E	6
F	8
G	2
H	0



- Use A* search with the strict expanded list version to find the shortest path from A to H, Strict Expanded List Rule: Once a node is expanded, it cannot be revisited. Show the final shortest path and its cost.

- All heuristics are consistent. If not, make them consistent.

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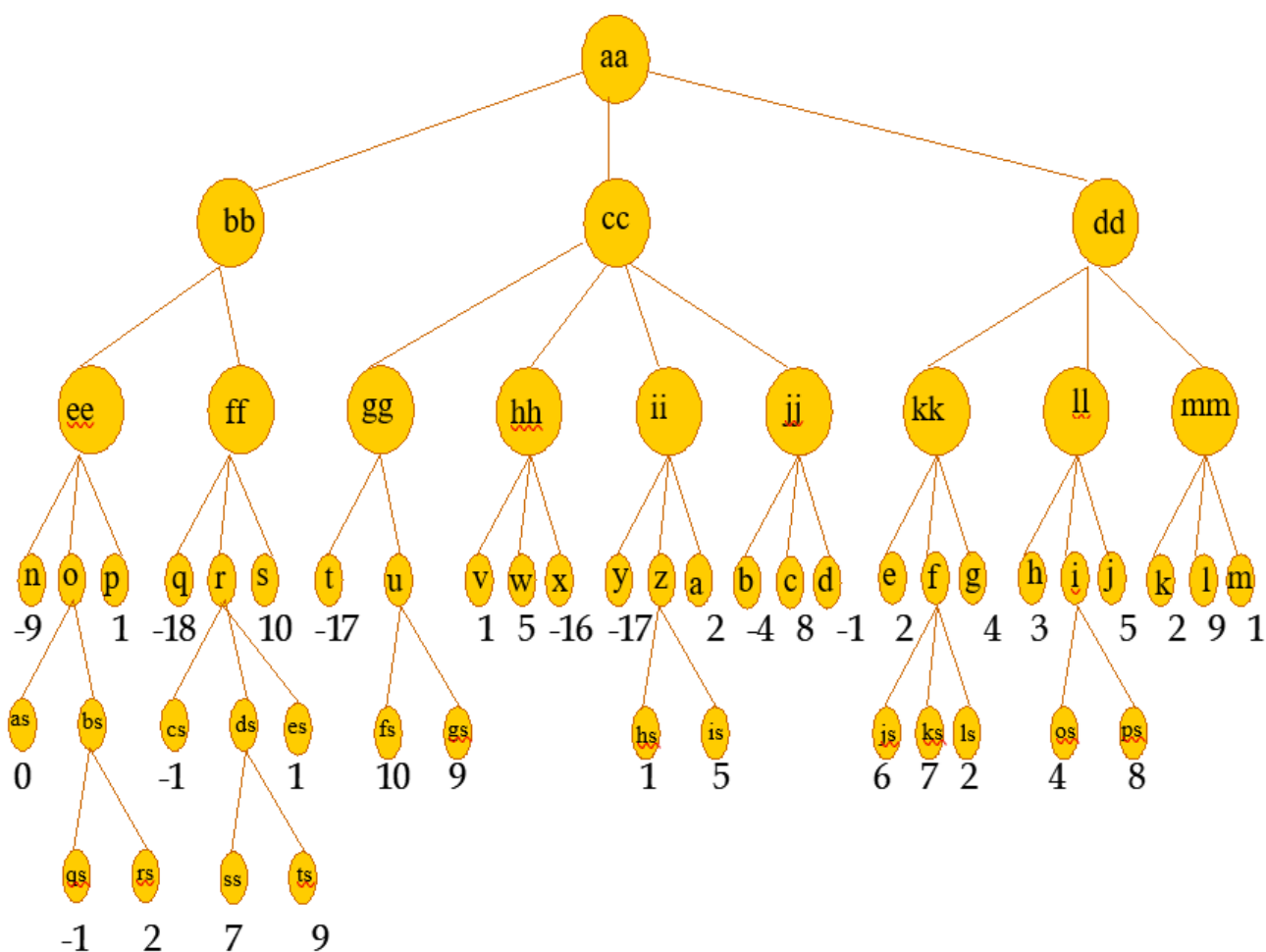
CLO [C3]: Knowledge of method of blind as well as informed search and ability to practically apply the corresponding techniques

Q3: Adversarial Searches

[6 + 4 = 10 Marks]

Use alpha beta pruning to find out which branches of the game tree are explored. If a branch is not explored then **cross it out**. If a branch is explored then put a **tick mark** against it. Don't leave it blank. Also, show the values of alpha and beta wherever necessary. Alpha is the score for max and beta is the score for min. **It is Min's turn to play at the root level.**

Part A: mentioned the values of alpha and beta at each node in the tree given below.



Part B: Write names of all nodes in left to right order which will get pruned

CLO [C3]: Knowledge of method of blind as well as informed search and ability to practically apply the corresponding techniques

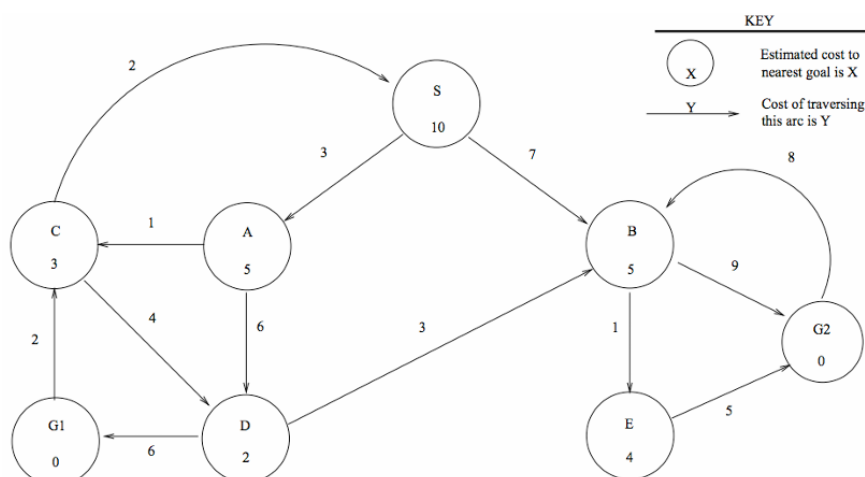
Q4: Uninformed Searches

[10 + 5 = 15 Marks]

Part A: Assume the following search graph, where S is the start node and G1 and G2 are goal nodes. Arcs are labeled with the cost of traversing them and the estimated cost to a goal is reported inside nodes. For each of the search strategies listed below, indicate which goal state is reached (if any) and list, in order, the states expanded. (Recall that a state is expanded when it is removed from the OPEN list.) When all else is equal, nodes should be expanded in alphabetical order. [10 marks]

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a) Breadth First Search [2 Marks]

Order of Expansion: _____

Final Path: _____

Final Cost: _____

b) Best First Search [2 Marks]

Order of Expansion: _____

Final Path: _____

Final Cost: _____

c) Iterative Deepening Search [4 Marks]

Order of Expansion: _____

Final Path: _____

Final Cost: _____

d) Uniform cost search [2 Marks]

Order of Expansion: _____

Final Path: _____

Final Cost: _____

Part B: Consider the following three search techniques: depth-first (with cycle checking), breath first, uniform-cost. Answer the following questions [5 marks]

1. For each of the three methods, state if they are optimal or not and under what conditions.

BFS	
DFS	
UCS	

2. If the search tree has constant branching factor of b and maximum depth m , with a shallowest solution being at depth s , what will the time and space complexity be for depth first (with cycle checking) and breadth first search.

DFS	
BFS	

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3. If b is large and no heuristic is available, which search method would you use and why? It need not be one of the 3 methods listed above.

[C2]: To introduce students with computational intelligence theory and techniques used to design and develop intelligent systems.

Q5: Propositional Logic

[5 + 5 = 10 Marks]

Part A: Consider, we have following Knowledge Base (KB). Convert it into CNF and use Resolution Theorem to accept or reject the Goal. [5 Marks]

1. The humidity is high or the sky is cloudy.
2. If the sky is cloudy, then it will rain.
3. If the humidity is high, then it is hot.
4. It is not hot.

Goal: it will rain.

Part B: Decide whether each of the following sentences is valid, unsatisfiable, or neither. Verify your answers using the equivalence rules. Justify your answer. [5 Marks]

1. $((P \rightarrow Q) \wedge (Q \rightarrow R)) \wedge \neg(P \rightarrow R)$

Neither, satisfiable

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2. $((P \rightarrow Q) \wedge (Q \rightarrow R)) \rightarrow (P \rightarrow R)$

valid

3. $(P \rightarrow Q) \leftrightarrow (\sim P \rightarrow \sim Q)$

Neither, satisfiable

4. $((\neg \neg C) \Rightarrow \neg D) \Rightarrow (D \Rightarrow \neg C)$

valid

5. If A is not satisfiable, then $\neg A$ is valid or not.

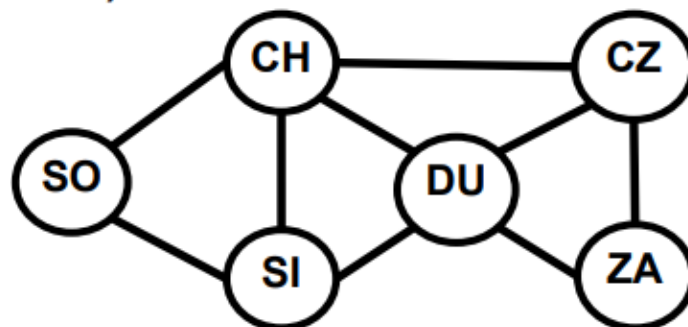
Valid

CLO [C2]: To introduce students with computational intelligence theory and techniques used to design and develop intelligent systems.

Q6: Constraint Specification Problem

[15 Marks]

You are a map-coloring robot assigned to color this Northwest Mexico map. Adjacent regions must be colored a different color (R=Red, B=Blue, G=Green). The constraint graph is shown.



FORWARD CHECKING: Cross out all values that would be eliminated by Forward Checking, after variable DU has just been assigned value G, as shown:

SO	CH	SI	DU	CZ	ZA

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Arc Consistency: SO = B and CZ = R have been assigned values, but no constraint propagation has been done. Cross out all values that would be eliminated by Arc Consistency.

SO	CH	SI	DU	CZ	ZA

MINIMUM-REMAINING-VALUES: Consider the assignment below. SI is assigned and constraint propagation has been done. List all unassigned variables that might be selected by the Minimum-Remaining-Values (MRV) Heuristic:

DEGREE HEURISTIC: Consider the assignment below. (It is the same assignment as in problem above.) SI is assigned and constraint propagation has been done. List all unassigned variables that might be selected by the Degree Heuristic

Constraint Propagation: perform arc-consistency process with no constraint propagation, eliminate all the values and fill in the following table with the final resulting values for all other variables.

SO	CH	SI	DU	CZ	ZA

CLO [C1]: To introduce the notion of intelligence and the so called artificiality associated with it, and how these can be modelled in computational system

Q7: Local Searches

[10 Marks]

Part A: In local search algorithm performing a hill-climbing gradient search, a state is found for which a step in any direction in feature space does not produce an increase in the score value, i.e., [3 Marks]

$$f(x_0) \leq f(x_{\text{current}}) : \{x_0 \in \text{Neighbors}(x_{\text{current}})\}$$

1. Have we found a global maximum? If not, what two other things we've discussed could it be?

2. What relatively simple additional method should we add to the basic hill-climbing algorithm in order to attempt to verify or disprove this?

Part B: Simulated Annealing

[4 Marks]

1. If we start the search with $T_0 = 1$ and cooling rate 1.1, will we be more or less likely to accept suboptimal s_0 over time as we iterate?

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2. Let $\text{succ}(x) = x'$, score $f(x) = f(x') = 1$ and $T = .9$. What is the probability we accept x' ?

Part C: Prove/Justify each of the following statements with suitable assumptions, or give a counterexample: [3 Marks]

1. Explain what it means when $\Delta E < 0$ Simulated Annealing. What does the algorithm do in this case?

2. In Simulated Annealing, what happens if the evaluation of a neighbor state is better than the current state?